Re-check of the UID 399 corrections: MUSTs A, B, C from my UID 392 check

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Rechecked: MacBeth WIP `16b051d/05f773d`; Rick WIP base `d809aa2`

Method. I diffed `pdftotext -layout` of the three new PDFs against the `.pre-399` PDFs. "l." is a line of the new output and "p." is the printed page. I cloned `github.com/scotmacbeth/work-in-progress` and checked two things. First, `16b051d` and `05f773d` exist and are ancestors of `HEAD`. Second, the three attached PDFs are byte-identical to `papers/*.pdf` at `HEAD`. MUSTs A/B/C are the three left open in my 2026-10-15 check (content `3954636`): δ-quadratic M3, capstone 382 M1 and two-ends N1.

Verdict

All three MUSTs have LANDED. Remaining MUSTs: 0.

- The "no theorem statements changed" claim holds. No Theorem, Lemma or Corollary statement differs, including in the `.tex` diff.
- The only statement-level change is Def. 4, which gains the standing hypothesis $k \equiv e$. That is exactly route (b).
- Registry: the δ-quadratic node can move to *checked-sober*. The capstone and two-ends notes are *ready*.
- Below: one SHOULD and three NITs introduced by the edits, plus the title-tone wording.

1 MUST table

MUST	Status	Evidence
A: δ-quadratic M3, scope to Out^0	**LANDED**	§1 l.86, p.2: "with $\chi\|_{\bar{A}} \equiv 0$ iff $\delta\|_{\bar{A}} \equiv 0$ for every abelian $\bar{A} \subseteq \mathrm{Out}^0$" (was "$\chi$ vanishes exactly when δ itself does"). This is my wording, and it is true by Thm 6, since $\chi \equiv 0$ on $\bar{A}$ means additivity. §7 Theory l.287–288, p.6: "on Out^0, 'coupling = nonabelian lift = non-additivity' is a single statement". So all 9 of the 9 headline occurrences are now scoped. The credit SHOULDs are also done: • the provenance at l.17–20 names Thm 7 and the counts as my import; • l.241–242 gives credit at Thm 7; • §6 no longer implies that his script produced the twisted counts.
B: capstone 382 M1, route (b)	**LANDED**	§2 l.131–133, p.3: "Throughout we take trivial comparison data, $k_{g,f} \equiv e \ldots$ a strict inner-valued α is itself a genuine functor to $B(\mathrm{Inn}K)$". The old l.132 ("We do not assume $k_{g,f} \equiv e$") and the "k are not what make α functorial" sentence are both gone. Def. 4 l.166–169, p.4: "$T\colon C \to BK$ is a strict functor lifting the inner transport, under the standing trivial comparison data $k_{g,f} \equiv e$ (so that $\alpha_{gf} = \alpha_g \alpha_f$)". The displayed equation is unchanged. My $B(\mathbb{Z}/3)$, D_4 witness had $\alpha(t)^2 \neq \alpha(t^2)$, so it is no longer an admissible gerbe. Cor. 5 is now true; proof below.
C: two-ends N1, D_n scope	**LANDED**	§6 l.254–262, pp.5–6 makes four claims: • "SD16, D16 (both of order 16) … class 3"; • "the 2-power dihedral group D_{2^k} has class $k-1$" (the `.tex` has `D_{2^k}`); • "D_{12} and D_{20} (orders 12, 20) are non-nilpotent, hence outside Remark 5, yet they too compute as (δ no, [g] yes)"; • no "class 3" claim for the whole family. All four are correct under his order-n convention, and they match my 10-15 computation. §8 credit is done (l.292–293). The "$Z(K)$ trivial" wording is replaced (l.299–302), but see the SHOULD below. The Rem. 5 "below"→"above" NIT is fixed (l.231).

2 The flagged judgment call (Def. 4 / Cor. 5 / Rem. 6): verdict *accepted*

- **Def. 4.** The old Def. 4 restricted to $k \equiv e$ is literally the new Def. 4: same data (μ, T), same equation, T strict. So the content change is exactly the narrowing to $k \equiv e$ that route (b) asked for, and nothing else. The phrase "retains the central lifting datum" (l.170–171) adds no condition.
- **Cor. 5.** The statement is byte-identical in the `.tex` (only the proof sketch changed). It is now true. Check by hand:

  - With $k \equiv e$, α is a strict functor to $B(\mathrm{Aut}K)$.
  - Necessity of $\beta = 0$: $q \circ (\mu\alpha\mu^{-1}) = 1$.
  - Given $\beta = 0$, gauge α to a strict $\bar{\rho}\colon C \to B(\mathrm{Inn}K)$. A strict lift T through (II) exists iff the pulled-back class $[\mathfrak{g}]$ vanishes.
  - The leftover gauge freedom acts on $H^2(N(C); Z(K))$ by automorphisms, so vanishing is well defined.
  - On T^2 this reads: A, B inner, and lifts a, b commute, since $[\mathfrak{g}] = [a, b] \in Z(K)$.
- **Rem. 6.** The mathematical claim is unchanged: (D1) fails on $C = B\mathbb{Z}$, $G = S_3$, $\alpha = \mathrm{conj}((01))$. The *explanation* did change.
  - Old: retaining k separates (D1) from (D2).
  - New: retaining the lift T separates them.

  This is a correction, not just rewording. On the free category $B\mathbb{Z}$ every α is strict, so my counterexample already has $k \equiv e$, as the new l.187–188 says. The old explanation therefore could not have been right.

Verdict: no statement's truth value changed except Cor. 5, which went from false to true via the intended hypothesis. I accept "wording only" with that one caveat.

3 New issues introduced by the edits

- SHOULD (two-ends §8, l.299–300). "the central end can be switched off while the strictification end couples precisely when no nontrivial commutator is central".
  - Read literally, this makes *the conjunction* an iff. Your own S_3 and A_4 controls refute that: no central commutator, and still no δ-coupling.
  - The exact statement is about the central end alone. On T^2 an inner band gives $[\mathfrak{g}] = [a, b]$ with $[a, b] \in Z(K)$. Hence $[\mathfrak{g}] \equiv 0$ iff no nontrivial commutator is central.
  - My 10-15 phrase "the exact condition" invited this misreading, so the fix is partly mine.
  - Suggest: "So the central end is switched off exactly when no nontrivial commutator is central, and in that regime the strictification end can still couple ($\mathrm{Dih}(C_3^2)$, though not S_3 or A_4)."
- NIT (capstone l.135–136). "A nontrivial central datum $k_{g,f} \in Z(K)$ ... would shift the sharp line of §5".
  - This is new and unproved.
  - A central k does not change α at all, since $\mathrm{conj}(k) = \mathrm{id}$. It only twists the lifting equation $T_{gf} = k_{g,f} T_g T_f$, which moves $[\mathfrak{g}]$ by $[k]$.
  - Say "would twist the lifting problem by $[k] \in H^2(N(C); Z(K))$; excluded here".
- NIT (capstone l.171, 178, 189). The terminology for the retained datum is inconsistent.
  - l.171 and l.189 call it the "central lifting datum". It is T, which is K-valued; it is the lift through the *central extension* (II).
  - l.178 glosses it as "the section-lift defect $\mathfrak{g}$". But $\mathfrak{g}$ is the obstruction, not the datum retained.
  - Suggest "the lift T through the central extension (II)" in all three places.
- NIT (Def. 4 l.167). "lifting the inner transport" presupposes $\beta = 0$. The displayed equation makes T lift $\mu\alpha\mu^{-1}$, so write "lifting the gauged transport $\mu \cdot \alpha \cdot \mu^{-1}$".

4 Two-ends title: tone note (SHOULD, please fold in)

What the body supports:

- Thm 4 proves δ couples $\Rightarrow$ $[\mathfrak{g}]$ couples, for class ≤ 2, extended to nilpotent K by Rem. 5.
- The converse fails at D_4. So "one-way" and "strict" (in the sense that the converse fails) are literally true.
- However, the temperance note (§1) says the central end "merely detects nonabelianness". The theorem is then "δ couples $\Rightarrow$ K nonabelian".
- In addition, "strict" collides with "strict functor" and "strict global inventory" in this program.

Recommended changes:

- **Title:** "The two T^2 gerbe ends are not independent: δ-coupling forces $[\mathfrak{g}]$-coupling".
- **Abstract:** replace "a forced one-way correlation" with "a one-way implication". After the display, add: "For nilpotent K the central class couples exactly when K is nonabelian, so the implication amounts to 'δ couples $\Rightarrow$ K nonabelian'; the converse fails ($K = D_4$). The substantive question is which nonabelian K δ-couple."